# Square–Hexagon Plane-Curve Period Certificate

Deductive exact reduction, reduced primitive, and replay payload

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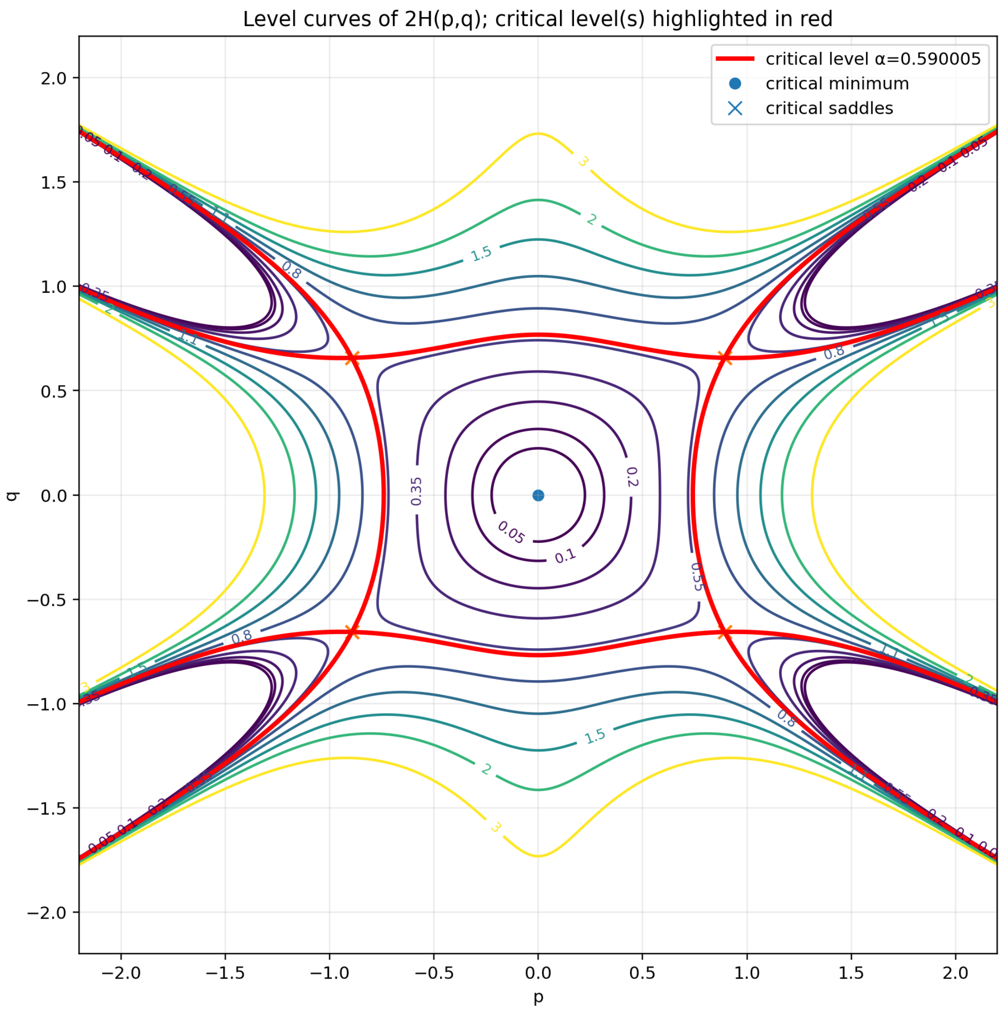


**Curve** $\alpha = 2H(p,q) = p^2 + q^2 - 2p^2q^2 + \frac{1}{4}p^2(p^2 - 3q^2)^2.$

**Pole** $\rho = (2H)_p = \frac{p}{2}(3p^4 - 12p^2q^2 + 9q^4 - 8q^2 + 4).$

**Period** $\omega = \frac{dq}{H_p} = \frac{2\,dq}{\rho}, \qquad T(\alpha) = \oint_{C_\alpha} \omega, \qquad T(0) = 2\pi.$

**Operator** $A_4 = \sum_{j=0}^{4} P_j(\alpha)\partial_\alpha^j.$

**Certificate** $\boxed{A_4\omega = d\Xi}, \qquad \Xi = \frac{V(\alpha,p,q)}{\rho^7}.$

The numerator is symmetry-compressed as

$$V = q\left(U_0(\alpha,q^2) + p^2U_1(\alpha,q^2) + p^4U_2(\alpha,q^2)\right),$$

with 40 nonzero coefficient polynomials and 514 expanded terms. The full numerator is embedded as machine-readable payload rather than printed as a wall of coefficients.

## Deductive closure

For an order-$r$ primitive $V/\rho^{2r-1}$, set the source weight $n = \deg_p + \deg_q$. A top-symbol minor of the exact-image map is

$$D_r(n) = 419904\,(n - (8r-3))(n - (8r-4))(n - (6r-3)).$$

Hence all source weights above $8r - 3$ vanish recursively, making the reduction finite and exhaustive. The exact ranks are

| $r$ | $n_{\max}$ | rows | exact cols | rank $C_r$ | rank$[C_r \mid W_r]$ | dim relations |
|---|---|---|---|---|---|---|
| 1 | 5 | 17 | 6 | 6 | 8 | 0 |
| 2 | 13 | 29 | 18 | 18 | 21 | 0 |
| 3 | 21 | 41 | 30 | 30 | 34 | 0 |
| 4 | 29 | 53 | 42 | 42 | 46 | 1 |

Thus the first exact cohomology relation occurs at order four and is unique. Fraction-free quotient reduction, without loading a guessed operator, returns exactly the $P_j$ printed below and exactly the stored numerator $V$.

## Reductive closure

The five operator polynomials have gcd 1. The 40 coefficient polynomials of $V$ have gcd 1 in $\mathbb{Q}[\alpha]$, and the numerical content of $V$ is 1/2. Therefore the integral payload is

$$2A_4\omega = d\left(\frac{2V}{\rho^7}\right).$$

Moreover $V$ is not divisible by $\rho$ in $\mathbb{Q}(\alpha)[p, q]/(2H - \alpha)$. Exact Gröbner reduction gives

$$\mathrm{NF}_{\langle 2H-\alpha,\rho\rangle}(V) = \frac{15qS_8(\alpha)}{2\alpha - 1}R(\alpha, q^2), \qquad R \neq 0.$$

Consequently the pole $\rho^7$ is reduced and cannot be replaced by any lower power. The degree-eight factor $S_8$ is the apparent factor in $P_4$.

## Exact order-four operator

The leading coefficient factors as

$$P_4 = 8\alpha(27\alpha^2 + 16)(486\alpha^3 - 792\alpha^2 + 632\alpha - 197)S_8(\alpha),$$

where

$$\begin{aligned} S_8 = {} & 2404239084\alpha^8 - 22448717379\alpha^7 + 16752988620\alpha^6 - 120418068744\alpha^5 \\ & - 163919968272\alpha^4 + 178534770624\alpha^3 - 49764209536\alpha^2 + 35868448768\alpha - 20307957760. \end{aligned}$$

$P_0(\alpha)$

$$\begin{aligned} P_0 = {} & 210322835068320\alpha^{10} - 3498482777898960\alpha^9 + 258867849968730\alpha^8 \\ & - 2558743410968856 0\alpha^7 - 15147053716188744 0\alpha^6 + 2362514686465536 00\alpha^5 \\ & - 5157299500910208 0\alpha^4 + 3607424411969664 0\alpha^3 - 7326500710640640 0\alpha^2 \\ & + 2739239496517632 0\alpha - 3161055263293440. \end{aligned}$$

$P_1(\alpha)$

$$\begin{aligned} P_1 = {} & 3654359259312060\alpha^{11} - 48803981756242707\alpha^{10} + 47241225871751484\alpha^9 \\ & - 376619237191140624\alpha^8 - 807894950107229280\alpha^7 + 1830637200177129888\alpha^6 \\ & - 1174507375213524096\alpha^5 + 618762464016620544\alpha^4 - 481098289601008128\alpha^3 \\ & + 279714899019497472\alpha^2 - 90774194832973824\alpha + 9669720119574528. \end{aligned}$$

$P_2(\alpha)$

$$\begin{aligned} P_2 = {} & 6265867794743700\alpha^{12} - 75876665783648193\alpha^{11} + 112112627078488392\alpha^{10} \\ & - 572523884171740728\alpha^9 - 469122498283346928\alpha^8 + 1816134511615957152\alpha^7 \\ & - 1881625776443915328\alpha^6 + 1326407688732674304\alpha^5 - 771727244978217984\alpha^4 \\ & + 409283435562258432\alpha^3 - 182117358126620672\alpha^2 + 50849032933474304\alpha \\ & - 8304120428888064. \end{aligned}$$

$P_3(\alpha)$

$$\begin{aligned} P_3 = {} & 2523874020819840\alpha^{13} - 28801428031403280\alpha^{12} + 53737798199249664\alpha^{11} \\ & - 212777881731573984\alpha^{10} - 8857742957395632\alpha^9 + 416970391270729248\alpha^8 \\ & - 643623914923349760\alpha^7 + 603478186349321088\alpha^6 - 409940704413324288\alpha^5 \\ & + 219849486323659776\alpha^4 - 105829863125549056\alpha^3 + 42068878165671936\alpha^2 \\ & - 11665912913985536\alpha + 1536256388628480. \end{aligned}$$

The expanded $P_4$ is omitted because the displayed factorization is exact.

## Replay and embedded payload

Run from the packet root:

```
python3 exact/verify_merged_certificate.py
python3 exact/derive_deductive_certificate.py
python3 exact/verify_reduced_primitive.py
python3 exact/verify_operator_ladder_400.py
python3 exact/verify_ore_relations.py
```

Expected markers are `MERGED_CERTIFICATE_PASS`, `DEDUCTIVE_CERTIFICATE_PASS`, `REDUCED_PRIMITIVE_PASS`, `OPERATOR_LADDER_400_PASS`, and `ORE_RELATIONS_PASS`.

The final PDF embeds the operator JSON, primitive JSON, integer-scaled primitive, compact numerator blocks, closure witness, replay scripts, report, and SHA-256 manifest as file attachments.

## Payload manifest

| SHA-256 prefix | attached file |
|---|---|
| 1bed0867cee2dd5f | exact/order4_operator.json |
| 6d778fae8e46ebbf | exact/order4_xi.json |
| fe8cdea8eaf2bd8e | exact/order4_xi_integer_scaled.json |
| 0852b6fee049c119 | closure/closure_witness.json |
| 86c24f44ead7e8bc | closure/primitive_compact.json |
| c5ea623a1f9da482 | exact/verify_merged_certificate.py |
| da84e1caca331780 | exact/derive_deductive_certificate.py |
| 884172c25150cf15 | exact/verify_reduced_primitive.py |
| 0ec10d2d79d4a396 | CLOSURE_REPORT.md |

**Scope.** The displayed identity, the order-four first-relation claim inside the exhaustive exact reduction, and the reduced-pole claim are closed. Coordinate changes that might beautify the operator and a conceptual explanation of the apparent factor remain open presentation problems.